

Rehab Exercise Monitoring System – Technical & Instruction Report

This document provides a detailed description of the three rehabilitation exercises currently implemented in the system. For each exercise, the report explains: 1. How the code works internally. 2. What the patient is instructed to do. 3. What kind of feedback the system provides.

1. Shoulder Flexion

Code Logic:

- Tracks shoulder–elbow–hip line to measure flexion angle.
- Uses smoothing to reduce jitter.
- Defines stages (down, moving, holding, done).
- Checks torso tilt and elbow bending.

Patient Instructions:

- Stand upright.
- Raise arm forward to overhead (~160–170°).
- Hold briefly, then lower slowly.

Feedback:

- Rep counted when overhead hold is correct.
- Warns for leaning or bent elbow.
- Visual guide + progress marker shown.

2. Farmer's Carry (Arm Raise + Carry)

Code Logic:

- Tracks static arm angle during walking.
- Monitors duration of correct posture.
- Checks arm height, torso tilt, elbow flexion.

Patient Instructions:

- Hold a weight, raise arm to ~90°.
- Walk slowly while keeping arm raised.
- Maintain upright posture.

Feedback:

- Encouragement if arm stays in correct zone.
- Warns if arm drops or torso leans.
- Visual overlay shows target angle and live progress.

3. Shoulder Abduction (Lateral Raise)

Code Logic:

- Tracks arm movement sideways (frontal plane).
- Uses shoulder–elbow–torso angles.
- Defines stages (down, moving, holding, done).
- Checks torso tilt and elbow bending.

Patient Instructions:

- Stand upright, arms at side.
- Raise arm sideways to overhead (~160°).
- Hold briefly, then lower slowly.

Feedback:

- Rep counted when hold is correct.
- Warns for leaning or bent elbow.
- Visual lateral path + progress shown.

Summary Comparison

Exercise	Plane of Motion	Patient Action	Focus	System Checks
Shoulder Flexion	Sagittal	Raise arm forward to overhead, hold, lower	Mobility + posture	Arm angle, elbow bend, torso tilt
Farmer's Carry	Static (walking)	Hold weight, arm raised at angle while walking	Endurance + posture	Arm height, stability, torso tilt
Shoulder Abduction	Frontal	Raise arm sideways to overhead, hold, lower	Mobility + posture	Arm angle, elbow bend, torso tilt